Multimodal Fusion Guide

This document summarizes the current bundle priorities and fusion logic used by the screening runtime.

1. Bundle Priorities

The current runtime preference order is:

- Text: text_transformer primary, text fallback
- Audio: audio primary, audio_spectrogram and audio_sequence secondary
- Image: image_dl primary, image fallback
- Comorbidity: classifier-chain ensemble blended with upstream text, audio, and image consensus

These priorities are intentional. The project keeps the strongest trained bundle in front for each modality and uses the other bu

2. Why The Priorities Matter

The system is designed to be robust in field use:

- strong modality bundles carry most of the final signal,
- auxiliary branches help when the primary bundle is uncertain,
- weaker bundle families do not overrule the stronger ones,
- the final score remains stable even when one modality is missing.

3. Audio Fusion

Audio uses:

- the main audio bundle as the primary scorer,
- audio_spectrogram as a secondary stabilizer,
- audio_sequence as a fallback stabilizer.

The runtime gives mood-heavy domains like depression, loneliness, anxiety, sleep disorder, and burnout slightly more room to

4. Image Fusion

Image uses:

- image_dl as the primary scorer,
- the classical image bundle as a stabilizer.

The DL model gets a slightly higher influence when both are available.

5. Text Fusion

Text uses:

- text_transformer when installed and available,
- the classical text bundle as fallback,
- small label-specific retuning for weaker text domains such as depression, loneliness, anxiety, substance abuse, sleep disorde

The stronger labels are left mostly unchanged.

6. Comorbidity Fusion

The comorbidity head combines:

- classifier-chain ensemble probabilities,
- upstream modality consensus from text, audio, and image,
- calibration and pairwise lift metadata.

This helps the weakest bundle benefit more from stabilized upstream predictions instead of relying only on deeper chain tuning

7. Related Files

- README.md (../README.md)
- docs/ARCHITECTURE.md (ARCHITECTURE.md)
- docs/MODEL_COMPARISON_MATRIX.md (MODEL_COMPARISON_MATRIX.md)
- docs/PROJECT_DOCUMENTATION.md (PROJECT_DOCUMENTATION.md)
- docs/DEEP_LEARNING_MODEL_GUIDE.md (DEEP_LEARNING_MODEL_GUIDE.md)